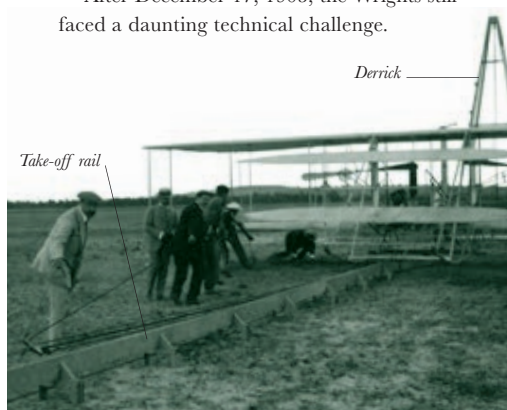


Langley's second attempt to fly his Aerodrome failed on December 8, and the way was now open for the Wrights. On their first attempt, on December 14, Wilbur could not control the machine, which came down heavily just after takeoff. But on Thursday, December 17, the goal of so many dreamers was finally attained (see pages 26–27).

Although the event did not go unreported – the Wrights themselves issued a press release – the public response was muted. The Langley Aerodrome fiasco had created a climate of skepticism, and most newspaper editors were inclined to dismiss claims of heavier-than-air flight out of hand. The attitude of the Wright brothers themselves did nothing to allay skepticism. In their January 1904 statement they concluded: “We do not feel ready at present to give out any pictures or detailed descriptions of the machine.” The Wrights had not originally pursued a policy of deliberate secrecy, but once they achieved powered flight, they were determined to stop anyone else from stealing their invention before they could profit from it.

After December 17, 1903, the Wrights still faced a daunting technical challenge.



ASSISTED TAKEOFF

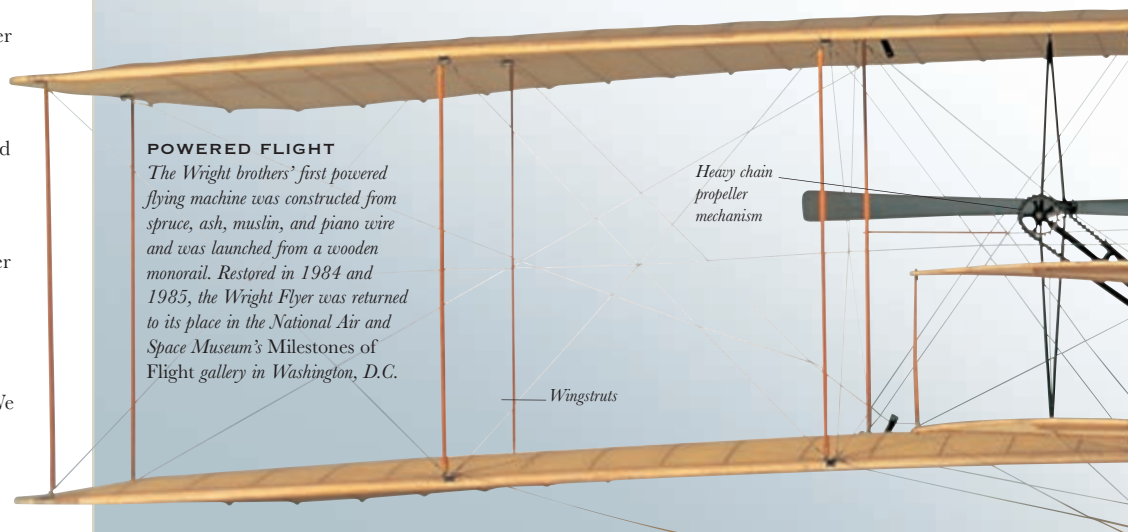
From 1904, the Wright brothers used the device shown above to assist takeoff. A weight, attached to the front of the airplane by a rope, was raised to the top of a derrick. When the airplane's engine had started and the pilot was ready, the weight was released and its fall jerked the machine along the rail.

Transferring their operations back to Dayton, they worked on building and testing improved models of their flying machine. The 1904 Flyer II had trouble getting off the ground under the very different weather conditions from Ohio. With the help of a catapult-assisted takeoff system, Flyer II proved capable of staying in the air for more than five minutes. However, between June and October 1905, in the much-improved Flyer III the Wrights made flights of up to 38 minutes' duration, covering more than 20 miles (30km) at a time. If anyone wanted to question whether the Wright

1903 Wright Flyer

POWERED FLIGHT

The Wright brothers' first powered flying machine was constructed from spruce, ash, muslin, and piano wire and was launched from a wooden monorail. Restored in 1984 and 1985, the Wright Flyer was returned to its place in the National Air and Space Museum's Milestones of Flight gallery in Washington, D.C.



EACH OF THE FLIGHTS made by the Wright brothers at Kitty Hawk in December 1903 was marked by instability, since the nose (and therefore the entire aircraft) would slowly bounce up and down. Sharp contact with the ground on the last flight broke the front elevator, ending that season's flying.

Between 1903 and 1908 the Wrights developed their original Flyer into a more robust and powerful machine, without making any fundamental changes to its configuration or control systems. All the Wright flying machines were controlled in pitch by the front elevator, in yaw by twin vertical rudders, and in roll by the twisting of the wing tips, known as wing warping.

Flying like cyclists, the Wrights kept the airplane balanced with continuous small adjustments of the

controls and leaning the machine into turns. This required considerable experience – a “feel” for flying that had to be learned. The trickiest feature was the front elevator, which tended to be overly sensitive. Any slight error of judgment could cause the aircraft to climb or dive alarmingly.



UNDER CONSTRUCTION

Orville Wright attaches wing-warping wires to a wing in the primitive hangar at Kill Devil Hills. Wing warping was devised by Wilbur to control the Flyer in roll. Twisting the wings to lift one side or the other allowed it not only to fly level, but also to make banked turns, rather like a bicycle cornering.

“I believe the new machine of the Wrights to be the most promising attempt at flight that has yet been made.”

OCTAVE CHANUTE, NOVEMBER 23, 1903

